

CO 1 ASSESSMENT TOOL- 1

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COURSE CODE: DSA0304

COURSE NAME: NATURAL LANGUAGE PROCESSOR

Q1. Resume Information Extraction

CODE:

```
import re
resume = """
Name: Arun Kumar
Email: arun@gmail.com
Mobile: 9876543210
Skills: Python, Java, SQL, Machine Learning, NLP
Experience: 3 years
"""

name = re.search(r"Name:\s*([A-Za-z ]+)", resume).group(1)
email = re.search(r"[\w.-]+@[ \w.-]+\.\w+", resume).group()
mobile = re.search(r"[6-9]\d{9}", resume).group()
experience = int(re.search(r"(\d+)\s*years?", resume).group(1))
skills = [
    skill for skill in
    ["Python", "Java", "SQL", "Machine Learning", "NLP"]
    if re.search(r"\b" + re.escape(skill) + r"\b", resume, re.I)]
print("Name:", name)
print("Email:", email)
print("Mobile:", mobile)
print("Skills:", ", ".join(skills))
print("Experience:", experience, "years")
if experience >= 2 and "Python" in skills:
    print("Eligibility: ELIGIBLE")
else:
    print("Eligibility: NOT ELIGIBLE")
```

```

import re
resume = """
Name: Arun Kumar
Email: arun@gmail.com
Mobile: 9876543210
Skills: Python, Java, SQL, Machine Learning, NLP
Experience: 3 years
"""

name = re.search(r"Name:\s*([A-Za-z ]+)", resume).group(1)
email = re.search(r"[\w.-]+@[\w.-]+\.\w+", resume).group()
mobile = re.search(r"[6-9]\d{9}", resume).group()
experience = int(re.search(r"(\d+)\s*years?", resume).group(1))
skills = [
    skill for skill in
    ["Python", "Java", "SQL", "Machine Learning", "NLP"]
    if re.search(r"\b" + re.escape(skill) + r"\b", resume, re.I)]
print("Name:", name)
print("Email:", email)
print("Mobile:", mobile)
print("Skills:", ", ".join(skills))
print("Experience:", experience, "years")
if experience >= 2 and "Python" in skills:
    print("Eligibility: ELIGIBLE")
else:
    print("Eligibility: NOT ELIGIBLE")

```

Name: Arun Kumar
 Email: arun@gmail.com
 Mobile: 9876543210
 Skills: Python, Java, SQL, Machine Learning, NLP
 Experience: 3 years
 Eligibility: ELIGIBLE

Q2. Product Search System

CODE:

```

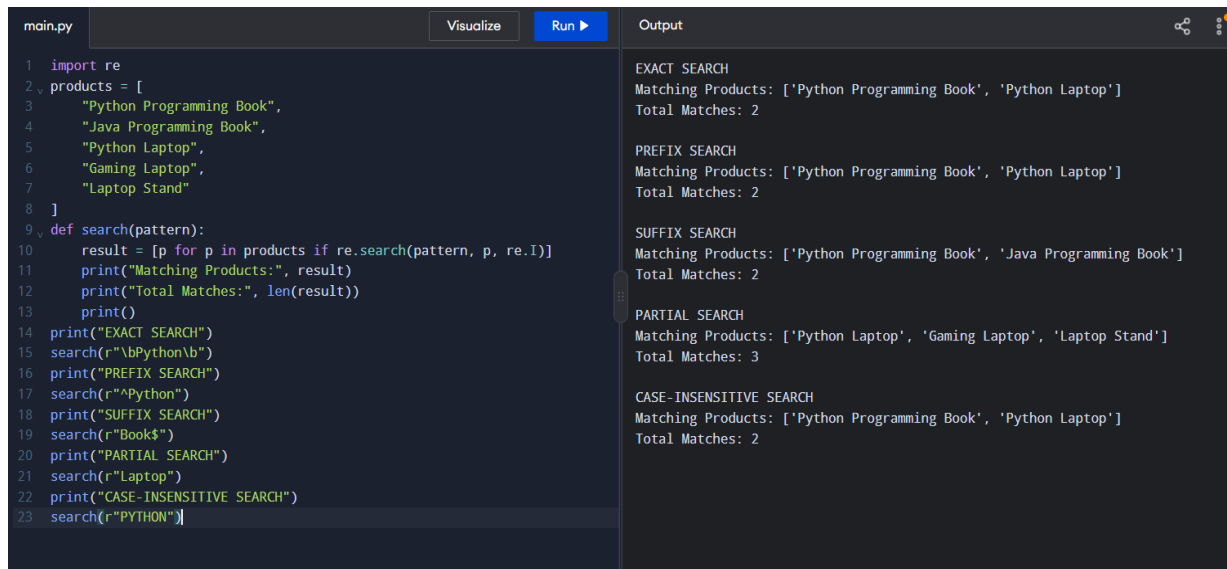
import re
products = [
    "Python Programming Book",
    "Java Programming Book",
    "Python Laptop",
    "Gaming Laptop",
    "Laptop Stand"
]

def search(pattern):
    result = [p for p in products if re.search(pattern, p, re.I)]
    print("Matching Products:", result)
    print("Total Matches:", len(result))
    print()

print("EXACT SEARCH")
search(r"\bPython\b")
print("PREFIX SEARCH")
search(r"^Python")
print("SUFFIX SEARCH")
search(r"Book$")
print("PARTIAL SEARCH")
search(r"Laptop")
print("CASE-INSENSITIVE SEARCH")

```

search(r"PYTHON")



The screenshot shows a Python IDE with a file named 'main.py'. The code defines a list of products and a search function. The search function iterates through five different regex patterns: 'PYTHON', 'Python', 'Python*', 'Python\$', and 'Laptop*'. For each pattern, it prints the matching products and the total number of matches. The output window on the right shows the results for each search.

```
1 import re
2 products = [
3     "Python Programming Book",
4     "Java Programming Book",
5     "Python Laptop",
6     "Gaming Laptop",
7     "Laptop Stand"
8 ]
9 def search(pattern):
10     result = [p for p in products if re.search(pattern, p, re.I)]
11     print("Matching Products:", result)
12     print("Total Matches:", len(result))
13     print()
14     print("EXACT SEARCH")
15     search(r"\bPython\b")
16     print("PREFIX SEARCH")
17     search(r"^Python")
18     print("SUFFIX SEARCH")
19     search(r"Book$")
20     print("PARTIAL SEARCH")
21     search(r"Laptop*")
22     print("CASE-INSENSITIVE SEARCH")
23     search(r"PYTHON")
```

Output:

```
EXACT SEARCH
Matching Products: ['Python Programming Book', 'Python Laptop']
Total Matches: 2

PREFIX SEARCH
Matching Products: ['Python Programming Book', 'Python Laptop']
Total Matches: 2

SUFFIX SEARCH
Matching Products: ['Python Programming Book', 'Java Programming Book']
Total Matches: 2

PARTIAL SEARCH
Matching Products: ['Python Laptop', 'Gaming Laptop', 'Laptop Stand']
Total Matches: 3

CASE-INSENSITIVE SEARCH
Matching Products: ['Python Programming Book', 'Python Laptop']
Total Matches: 2
```

Q3. University Registration Validation

CODE:

```
import re
register_no = "23AD123"
email = "23ad123@saveetha.com"
course = "DSA03"
semester = "Semester 5"
mobile = "9876543210"
valid = True
if re.fullmatch(r"\d{2}AD\d{3}", register_no):
    print("Register Number: VALID")
else:
    print("Register Number: INVALID")
    valid = False
if re.fullmatch(r"\d{2}ad\d{3}@saveetha\.com", email, re.I):
    print("Email: VALID")
else:
    print("Email: INVALID")
    valid = False
if re.fullmatch(r"[A-Z]{3}\d{2}", course):
    print("Course Code: VALID")
```

```
else:
    print("Course Code: INVALID")
    valid = False
if re.fullmatch(r"Semester [1-8]", semester, re.I):
    print("Semester: VALID")
else:
    print("Semester: INVALID")
    valid = False
if re.fullmatch(r"[6-9]\d{9}", mobile):
    print("Mobile: VALID")
else:
    print("Mobile: INVALID")
    valid = False
print("Registration:",
      "SUCCESSFUL" if valid else "UNSUCCESSFUL")
```

main.py	Visualize	Run ▶	Output
<pre>1 import re 2 register_no = "23AD123" 3 email = "23ad123@saveetha.com" 4 course = "DSA03" 5 semester = "Semester 5" 6 mobile = "9876543210" 7 valid = True 8 if re.fullmatch(r"\d{2}AD\d{3}", register_no): 9 print("Register Number: VALID") 10 else: 11 print("Register Number: INVALID") 12 valid = False 13 if re.fullmatch(r"\d{2}ad\d{3}@saveetha\.com", email, re.I): 14 print("Email: VALID") 15 else: 16 print("Email: INVALID") 17 valid = False 18 if re.fullmatch(r"[A-Z]{3}\d{2}", course): 19 print("Course Code: VALID") 20 else: 21 print("Course Code: INVALID") 22 valid = False 23 if re.fullmatch(r"Semester [1-8]", semester, re.I): 24 print("Semester: VALID") 25 else: 26 print("Semester: INVALID") 27 valid = False 28 if re.fullmatch(r"[6-9]\d{9}", mobile): 29 print("Mobile: VALID") 30 else: 31 print("Mobile: INVALID") 32 valid = False 33 print("Registration:", 34 "SUCCESSFUL" if valid else "UNSUCCESSFUL")</pre>			<pre>Register Number: VALID Email: VALID Course Code: VALID Semester: VALID Mobile: VALID Registration: SUCCESSFUL</pre>